Textual Analysis of Hillary Clinton's Emails

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Contents

Introduction

Hillary Clinton made headlines prior to Election Day 2016, but not in the way one would have expected. Rather than a new poll announcing her predicted victory, or for a speech made on the campaign trail, Clinton made headlines for her continued private-email controversy.

During her time as Secretary of State, Hillary Clinton never activated her state.gov email address. Instead, she used a personal account that was connected to a private server located in her New York home, for all of her communications. Much of the media's coverage of the issue was centered around whether Clinton broke the law and whether classified information had been compromised (BBC 2017, Bradner 2016, Theoharis 2015, Thiessen 2018).

Arguments on both sides of the political spectrum were apparent throughout the media's coverage. Some people claim that Clinton's predecessors, like Colin Powell, also used a private email account for many communications while in office. Furthermore, supporters cite that Clinton was known for lacking technology skills, a 'technophobe,' so she used the one email account on her Blackberry for convenience. However, others argue that the actions of the predecessors are irrelevant since Clinton's actions were out of line with current State Department guidelines. Many suspected there was an ulterior motive behind the private account as well, as a private email allows for control over what information enters the public domain. Regardless, there is a general consensus from both sides that the FBI's highly publicized email investigation cost Hillary Clinton the election, as the majority of the public no longer trusted her (BBC 2017, Bradner 2016, Graff 2016, Theoharis 2015, Thiessen 2018, Zurcher 2016). Despite widespread coverage into whether or not Clinton broke the law and what the effects of her actions were, there remains one underlying question: what exactly was in those emails? Two opposing answers are consistently seen in the media: most of the emails were innocuous, or most contained classified information, specifically regarding incriminating information about Benghazi (BBC 2017, Bradner 2016, Thiessen 2018). Yet, these views are mainly speculations as no effort has been made to sort the emails in a way which is digestible to the public. Thus, for my study, I aim to use textual analysis methods to begin to provide meaningful insight into the contents of the Clinton emails. Based on the media and political elites' claims, I predict that I will primarily find innocent topics like errands and some references to Benghazi and Libya.

Description of the Data

The State Department, in conjunction with the FBI, released several thousand emails from Clinton's private server. These emails, approximately 7,000 in number, are accessible in various formats and sites online. For my study, I utilized a downloadable text file which contained all the emails together from the non-profit library, Internet Archive. Large text files, especially in the form of emails, are extremely messy datasets. A substantial amount of time must be taken to preprocess and clean the file before analysis can occur. Stop words, or common English words with very little meaning, are first removed. For this dataset, I removed an additional list of similar words and a few other phrases unique to the emails. There is also a header at the top of each new email in the file, placed in by the State Department pertaining to the case number and information. Words used in these headers were removed along with the email start words: to, from, and subject. The cleaned text file was then used as the dataset for the duration of the analysis done.

Methods

Textual analysis is a growing field for political scientists. This is no surprise as most of our daily communication, and that of political elites, is now conducted through written, electronic forms, like email. The sheer volume of these correspondences makes it nearly impossible for meaningful studies to be conducted. The time and money it takes to hire hand-coders to read political texts means only very well-funded studies can come to fruition. Textual analysis provides the perfect solution to this problem as it allows political scientists to quickly interpret and draw substantive conclusions about the data (Denny & Spirling 2018, Grimmer & Stewart 2013).

Chosen Analysis Tools

For this study, I utilized three main textual analysis tools: word clouds, ngrams, and topic models. Word clouds clearly display the key components of the text by displaying high frequency words as larger and in the center of the plot, and with the less frequent words smaller around the outside. The method of analysis is based on a relatively simple calculation that keeps track of how frequent each word is in the corpus. These word-frequency pairs are then essential when determining the word cloud visualizations. Ngrams generate a list of words that are used in conjunction with one another. This can be crucial in ensuring certain words are given the proper context, as will be explored later. I chose to run this analysis for word groupings of two and three, known as bigrams and trigrams. Finally, topic modeling takes a number of categories as an input, determines what these categories should be, and then assigns words to the generated categories. Overall, word clouds allow for a mass visualization and understanding of the various items Clinton discusses in her emails, while the ngrams and topic models delve deeper to provide more context into how exactly Clinton talks about them. The majority of my textual analysis has its basis in unsupervised learning. Textual analysis is used to describe the content and structure of various messages within an electronic version of a text. Unsupervised learning allows for the algorithm to draw inferences without any predetermined parameters. With such a large corpus, as are most political texts, there truly are no a priori expectations which are certain enough to guide analysis. So, unsupervised learning is a common tool used by political scientists (Denny & Spirling 2018, Grimmer & Stewart 2013).

Limitations of the Methods

There are several limitations of the methods, but perhaps the largest in this study is the complexity of the English language itself. Two sentences with the same words can have very different meanings based on the order of those words, or even the grammar. Thus, conducting a study which serves to summarize context of a corpus based on individual words will never be perfect. To combat this, a crucial step in text analysis is critically examining the output of the model to ensure it makes sense both intuitively and based on theory (Denny & Spirling 2018, Grimmer & Stewart 2013). An additional limitation rests with the Latent Dirichlet Allocation algorithm, or LDA. LDA is the underlying algorithm of topic modelling that simultaneously determines the mixture of words in each topic, as well as the mixture of topics in the text. However, topic modeling requires you to set the number of groups, k, as an argument. With no a priori expectations, there is no concrete method for determining what the right k value is. While research into validation methods is a growing field, I followed common practice of testing several k values to compare (Silge & Robinson 2019).

Results

When initially running the word cloud and ngram analysis, the predominate outcomes are the names of Clinton's closest contacts. Chief of Staff Cheryl Mills, policy advisor Jacob Sullivan, and top aide Huma Abedin appear most prominently in the word cloud as well as clutter the ngram analysis. This shows that the majority of her correspondence is with her colleagues, rather than with her husband or other family members. Thus, we are able to extrapolate that most of the emails are not solely about non-political errands, as some claim.

```
# visualize most frequently used terms
set.seed(2345)


wordcloud(names(frequency), frequency,
          min.freq = 1, # terms used at least once
          max.words = 100, # 300 most frequently used terms
          random.order = FALSE, # centers cloud by frequency, > = center
          main = "Title",
          colors = brewer.pal(8, "Dark2"))
```

```
## Warning in wordcloud(names(frequency), frequency, min.freq = 1, max.words =
## 100, : afghanistan could not be fit on page. It will not be plotted.

## Warning in wordcloud(names(frequency), frequency, min.freq = 1, max.words =
## 100, : pakistan could not be fit on page. It will not be plotted.

## Warning in wordcloud(names(frequency), frequency, min.freq = 1, max.words =
## 100, : diplomacy could not be fit on page. It will not be plotted.
```


group
former
leaders
issues
administration
richard
great
military
minister
statement
percent
washington
issue
lauren
health
year
policy
back
including
people
security
take
countries
human
talk
working
june
government
tomorrow
party
called
staff
fyi
cheryl
public
report
political
women
told
american
years
abedin
huma
jacob
good
bill
say
world
iran
israel
mills
call
united
right
house
haiti
peace
secretary
time
much
says
states
press
support
sullivan
work
going
national
obama
hillary
rights
president
deal
need
israeli
media
please
plan
foreign
meeting
help
august
news
white
another
international
many
speech
global
process
senate
david
country
saturday
times
development
conference

Bigram	Frequency
jacob j	2307
sullivan jacob	2172
cheryl d	2154
mills cheryl	2152
abedin huma	1908
united states	999
huma abedin	652
white house	567
prime minister	493
human rights	474
verma richard	374
richard r	366
slaughter annemarie	357
foreign policy	324
lona j	309
valmoro lona	308
mchale judith	303
health care	296
en route	295
middle east	285
lauren ltjilotylc	274
president obama	266
northern ireland	258
national security	250
d Sullivan	244
obama administration	233
public diplomacy	232
foreign minister	215
j mills	203
united nations	203
washington dc	201
j abedin	192
assistant secretary	186
barack Obama	182
call sheet	182
cheryl mills	176
tea party	173
west bank	171
saudi arabia	170
muscatine lissa	169
per cent	167

Figure 1: “Bigrams With Names”

As is known, names are included in the start of every email when one addresses the person he/she is talking to, and may be referenced in the body of the text as well. Since the names were so prevalent, they crowded out the rest of the substantive data regarding the topics Clinton discussed. While acknowledging that Clinton’s most frequent contacts are important. I removed them and then ran the models again to highlight the other terms that dealt more directly with substantive messages. I show both models for comparison. However, the models ran without the names allows me to explore my research question in greater depth. I also chose to only run the topic models with this no-name dataset for this same reason.

```
set.seed(2345)


wordcloud(names(frequency_nonames), frequency_nonames,
          min.freq = 1, # terms used at least once
          max.words = 100, # 300 most frequently used terms
          random.order = FALSE, # centers cloud by frequency, > = center
          rot.per = 0.30, # sets proportion of words oriented horizontally
          main = "Title",
          colors = brewer.pal(8, "Dark2"))
```

Trigram	Frequency
sullivan jacob j	2172
mills cheryl d	2147
verma richard r	366
valmoro lona j	308
d sullivan jacob	244
j mills cheryl	203
j abedin huma	191
jacob j abedin	175
cheryl d sullivan	169
mashabane call update	163
jacob j mills	149
crowley philip j	132
verveer melanne s	132
huma sullivan jacob	127
lew jacob j	127
abedin huma sullivan	126
abedin huma saturday	123
reason d declassify	123
classified class confidential	121
confidential reason d	121
feltman jeffrey d	121
class confidential reason	120
d abedin huma	112
balderston kris m	106
lona j ltvalmorou	106
jacob j saturday	101
cheryl d abedin	93
lauren ltjilotylc abedin	86
ltjilotylc abedin huma	86
health care reform	82
burns william j	80
koh harold hongju	80
lona valmoro special	80
special assistant secretary	77
brava issue statement	75
bravo brava issue	75
huma mashabane call	75
mubarak call sheet	75
abedin huma mashabane	74
j sullivan jacob	74

Figure 2: “Trigrams With Names”


call
president
secretary
government
obama
people
time
meeting
american
states
israel
security
public
haiti
foreign
work
united
house
international
support
political
policy
world
party
many
year
years
back
washington
minister
good
need
much
must
might
election
officials
morning
asked
come
talks
diplomacy
times
pakistan
talk
press
white
leaders
senate
rights
right
media
says
plan
saturday
statement
health
israeli
hillary
take
country
great
august
help
global
war
report
national
fyi
women
issue
called
working
told
deal
administration
human
bill
say
staff
tomorrow
issues
iran
percent
speech
news
group
june
peace
going
countries
development
please
process
another
military
including
afghanistan
former
conference

Bigram - no names	Frequency
united states	999
white house	567
prime minister	493
human rights	474
slaughter annemarie	357
foreign policy	324
health care	296
en route	295
middle east	285
president obama	266
northern ireland	258
national security	250
obama administration	233
public diplomacy	232
foreign minister	215
united nations	203
washington dc	201
assistant secretary	186
barack obama	182
call sheet	182
tea party	173
west bank	171
saudi arabia	170
muscatine lissa	169
per cent	167
call update	163
mashabane call	163
public affairs	159
private residence	154
secretary hillary	152
around world	146
melanne s	141
food security	140
peace process	139
reines philippe	136
verveer melanne	135
security council	134
chief staff	132
foreign affairs	128

Figure 3: "Bigrams No Names"

These models display a much more clear idea of the various topics Clinton talked about in her emails. Words like 'Israel,' 'security,' 'haiti,' 'call,' and 'meeting' now appear larger and closer to the center. Ngram analysis also gives a much more meaningful output without the clutter of connecting each persons name. 'West Bank,' 'foreign policy,' and 'peace process' now appear closer to the top and are followed with various other meaningful pairs.

The series of topic models did not provide as clear of results as I had expected. Included in the Appendix, models were run with up to 30 topics to help distinguish the variety of political events and issues that Clinton could have discussed. Originally, I had hoped that the topic models would be able to classify words into topics surrounding varying policy issues, providing context for the individual and groupings of words seen in the word cloud and ngram analysis, and thus enabling an understanding of which words she used to describe each issue. However, topics were very similar to each other in word composition and did not provide much valuable insight.

Discussion and Conclusion

Further clarification on the probable historical context of these words can also be formed by utilizing public knowledge and news sources. Although this information will not tell us the specifics about what was explicitly said in Clinton's emails, it will provide a general sense as to why some words were so frequent.

The potential benefits of utilizing historical context can be highlighted in a number of words determined by the models. 'Haiti' frequently appears in Clinton's emails, and from a historical standpoint, we can assume that it is most likely in relation to the 2010 earthquake. It can also be concluded that the ngrams 'health care' and 'health care reform' are associated with the the Affordable Care Act. Oddly, the name of the act does not appear as a frequent term, as one might have expected. Implementation of historical context gets

Trigrams – no names	Frequency
mashabane call update	163
verveer melanne s	132
reason d declassify	123
classified class confidential	121
confidential reason d	121
feltman jeffrey d	121
class confidential reason	120
balderston kris m	106
health care reform	82
burns william j	80
koh harold hongju	80
special assistant secretary	77
brava issue statement	75
bravo brava issue	75
mubarak call sheet	75
president barack obama	72
george w bush	69
muscatine lissa ltmuscatinel	69
hanley monica r	68
call thru ops	65
jacobs janice l	65
drive time minutes	64
room th floor	64
slaughter annemarie j	64
depart en route	63
rodriguez miguel e	63
melanne s ltverveerms	62
public diplomacy public	57
staff meeting secretary	57
en route private	56
gonzalez juan s	56
route private residence	56
via cingular xpress	56
aid dollars buy	55
billions aid dollars	55
buy little goodwill	55
call friends haiti	55
cingular xpress mail	55
conference call friends	55
dollars buy little	55

Figure 4: “Trigrams No Names”

more difficult concerning terms like 'foreign policy' and 'peace process,' as a variety of specific issues fit into these broad policy topics. For instance, there is confusion about whether 'peace process' refers to the Middle East, Israel and Palestine on the West Bank, or to Northern Ireland since all are issues Clinton handled that appear in the models.

Along with insights from public knowledge of history, the importance of utilizing multiple models can also be seen in the study. Running multiple models gives crucial context to singular words, which can be seen in the case of the word 'food.' When I initially saw the word 'food' after running the word cloud, I assumed it was an indicator that Clinton was discussing a grocery errand. However, after the bigram analysis, it became evident that this was not the case. The word 'food' became 'food security,' clearly a political issue which probably relates to Clinton's Global Food and Hunger Initiative.

Much more analysis still needs to be done in order to fully understand the content of Clinton's emails and political risks associated with them. Despite hours of preprocessing, more steps can be taken to further clean the data. This is evident in the word clouds and as I had to manually delete several rows out of the ngram tables because they contained nonsensical groups like ascii characters. Additionally, most of the topic models did not cater to a greater understanding of the policy issues within the emails. During Clinton's four years in office using this email, one can imagine the plethora of issues she discussed. Further research should continue to look into validation methods for choosing the correct number of topics. This, along with cleaner data, may produce beneficial topic models which provide a better summary of unique policy issues.

My hypothesis prior to the study was that the emails would mainly contain either innocent references to errands or repeated mentions of the Benghazi incident. As displayed, neither of these topics constituted the majority of Hillary Clinton's emails. Instead, the most common references were to a variety of domestic issues as well as international issues that were not Benghazi related. The divide between what I expected based on media coverage and what I found poses a broader problem of the large impact the media has on issue framing. Especially when the proper information is not available to the media. Given the current trend in 'fake news' research, consideration should be placed on this less extreme version of misinformation, in the form of speculative reporting, as well.

References

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Appendix

Preprocessing Code

```
library(tm)
library(ggplot2)
```

```r
library(wordcloud)
library(topicmodels)
library(tidytext)
library(dplyr)


setwd("C:/Users/Emma/Documents/Senior Seminar Paper")
# set working directory
email_txt <- file.path("emails")
# chose folder with email text file

# create raw corpus, which will be preprocessed in a moment
email <- VCorpus(DirSource(email_txt))


# cleaning/ preprocessing

email <- tm_map(email, tolower)
email <- tm_map(email, removeWords, stopwords("english"))
email <- tm_map(email, removePunctuation)
email <- tm_map(email, removeNumbers)
email <- tm_map(email, stripWhitespace)
email <- tm_map(email, PlainTextDocument)

for (char in seq(email)){
  email[[char]] <- gsub("stategovgt", " ", email[[char]])
  email[[char]] <- gsub("â€", " ", email[[char]])
  email[[char]] <- gsub("@", " ", email[[char]])
  email[[char]] <- gsub("ltsullivanjj", " ", email[[char]])
  email[[char]] <- gsub("ltmillscd", " ", email[[char]])
  email[[char]] <- gsub("lthdrclint", " ", email[[char]])
  email[[char]] <- gsub("ltabedinh", " ", email[[char]])
  email[[char]] <- gsub("mailcomgt", " ", email[[char]])
  email[[char]] <- gsub("lthdr", " ", email[[char]])
  email[[char]] <- gsub("clintonemailcomgt", " ", email[[char]])
}


otherwords <- c("can","well","let","best","like","now","last","sun","still","dont","see","next",
                "now","mar","also","case","subject","unclassified","department","can","state",
                "two","one","know","doc","date","will","made","new",
                "just","think","want","sent","release","part","january","july","get","original",
                "message","said", "first","sat","sure","whether","even","fri","jan","yet",
                "even","feb","day","day", "got","tue","may", "thu","wed", "mon", "seen",
                "oct", "stategov", "email", "tuesday","december",
                "clintone", "sunday","clinton","friday","really","sid","pis","else","way","monday","marc
                "mailtohdr","sep","nov","end","aug","lthdr","since", "office", "dec", "h","c","cc","b",
                "fw","f", "gt","pm","re","thx", "make","thanks", "make","known", "today","thursday",
                "forward","wednesday", "week", "full", "jiloty", "lthrodclintone", "us", "ltvermarr",
                "ltmchaleja", "ltslaughtera", "das","gis","dos", "s", "april", "february")
```

```
email <- tm_map(email, removeWords, otherwords)


email <- tm_map(email, stripWhitespace)
email <- tm_map(email, PlainTextDocument)
```

DTM and Frequency Code, and Repeated with Name Removal

```
# dtm: each document is a row, and each term is a column
dtm <- DocumentTermMatrix(email)

frequency <- sort(colSums(as.matrix(dtm)), decreasing=TRUE)
# adding up the number of times each term is used, and sorting based on frequency of usage
```

```
names <- c("cheryl","huma","abedin","mills","jacob","sullivan","david",
           "verma","lauren","valmoro", "lona", "mchale","judith", "richard", "crowley",
           "philip")
email_nonames <- tm_map(email, removeWords, names)
email_nonames <- tm_map(email_nonames, PlainTextDocument)



# create new dtm and calculate new frequencies

dtm_nonames <- DocumentTermMatrix(email_nonames)

frequency_nonames <- sort(colSums(as.matrix(dtm_nonames)), decreasing=TRUE)
```

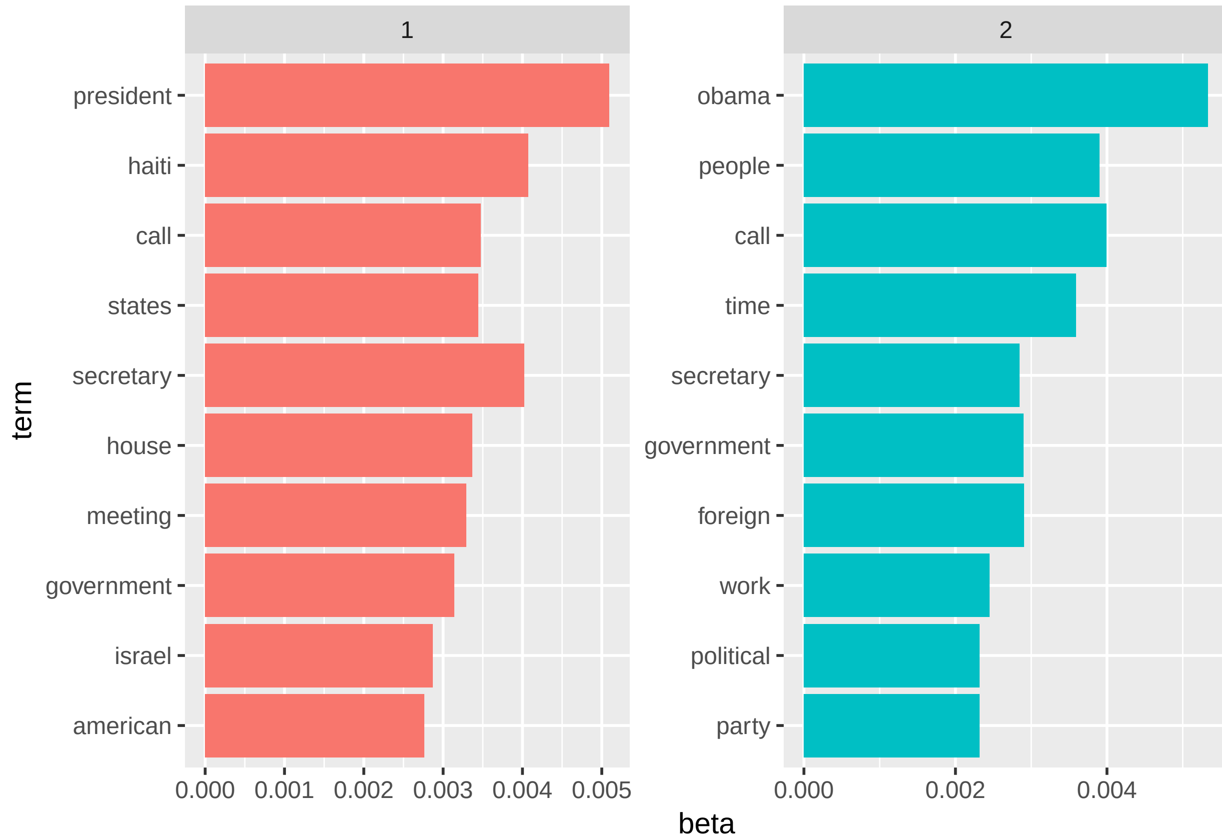
Ngrams Code

```
BigramTokenizer <- function(x)
  unlist(lapply(ngrams(words(x), 2), paste, collapse = " "), use.names = FALSE)
tdm.bigram <- TermDocumentMatrix(email, control = list(tokenize = BigramTokenizer))
tdm.bigram

freq <- sort(rowSums(as.matrix(tdm.bigram)),decreasing = TRUE)
freq.df <- data.frame(word=names(freq), freq=freq)
head(freq.df, 50)




TrigramTokenizer <- function(x)
  unlist(lapply(ngrams(words(x), 3), paste, collapse = " "), use.names = FALSE)
tdm.trigram <- TermDocumentMatrix(email, control = list(tokenize = TrigramTokenizer))


freq <- sort(rowSums(as.matrix(tdm.trigram)),decreasing = TRUE)
freq.df <- data.frame(word=names(freq), freq=freq)
head(freq.df, 50)
```

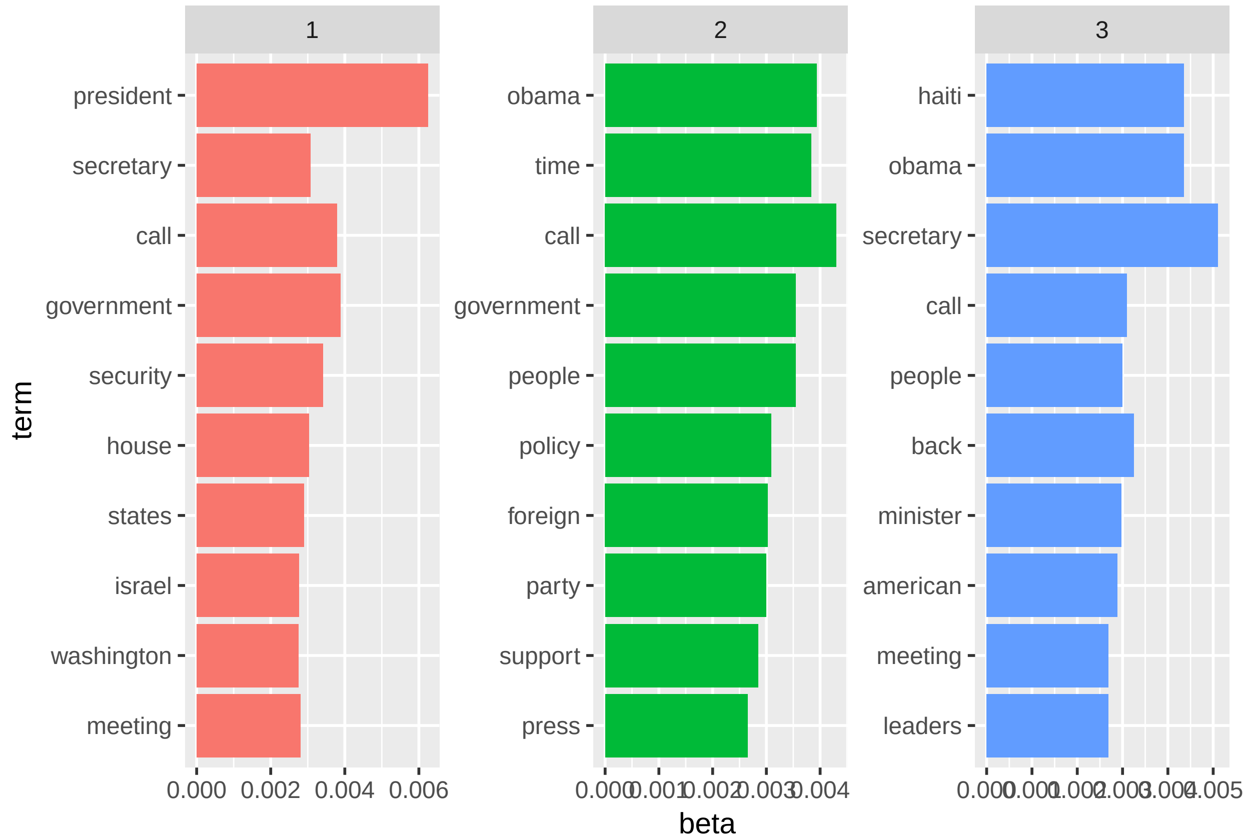
```r
BigramTokenizer <- function(x)
  unlist(lapply(ngrams(words(x), 2), paste, collapse = " "), use.names = FALSE)
tdm.bigram <- TermDocumentMatrix(email_nonames, control = list(tokenize = BigramTokenizer))


freq_nonames <- sort(rowSums(as.matrix(tdm.bigram)),decreasing = TRUE)
freq.df <- data.frame(word=names(freq_nonames), freq=freq_nonames)
head(freq.df, 50)




TrigramTokenizer <- function(x)
  unlist(lapply(ngrams(words(x), 3), paste, collapse = " "), use.names = FALSE)
tdm.trigram <- TermDocumentMatrix(email_nonames, control = list(tokenize = TrigramTokenizer))


freq_nonames <- sort(rowSums(as.matrix(tdm.trigram)),decreasing = TRUE)
freq.df <- data.frame(word=names(freq_nonames), freq=freq_nonames)
head(freq.df, 50)
```

Topic Models

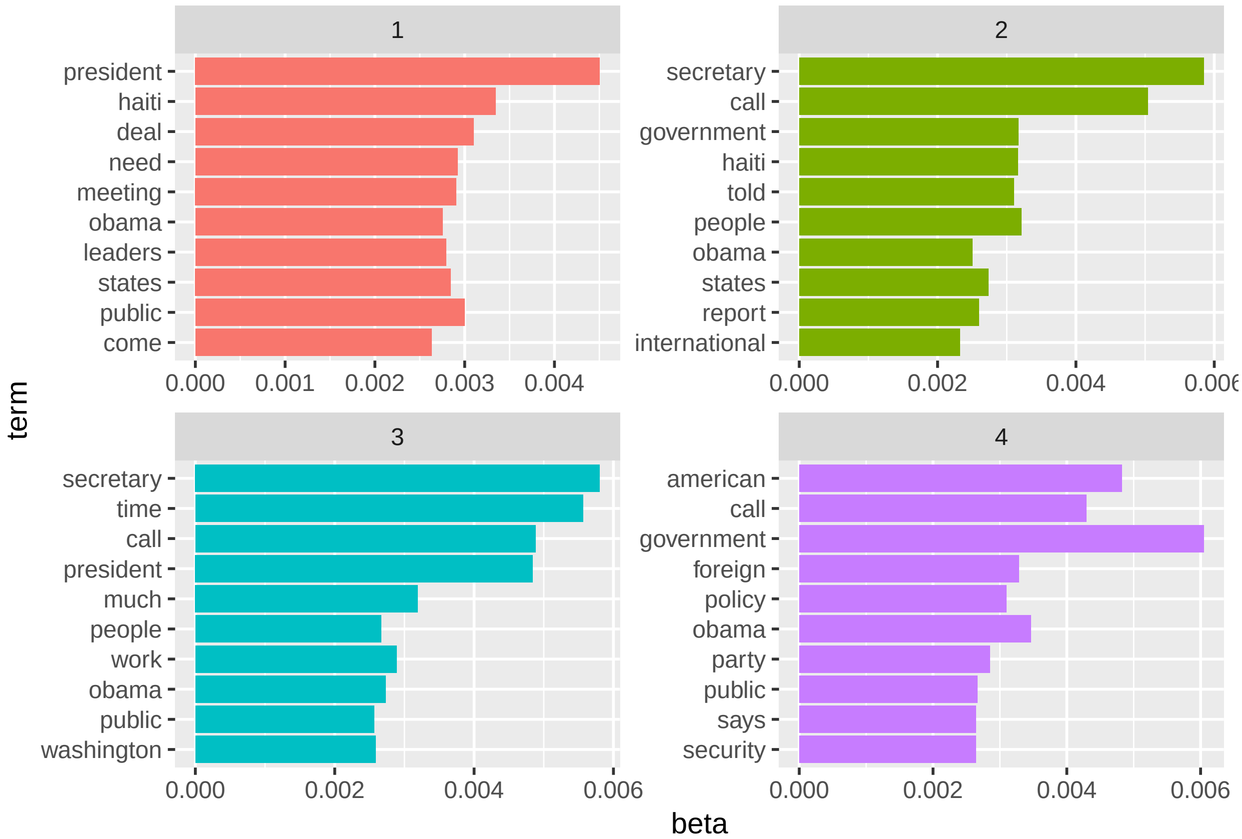
```r
# 2 groups

email_nonames_2topic <- LDA(dtm_nonames, k=2 )

topics2 <- tidy(email_nonames_2topic, matrix= "beta")

top_terms <- topics2 %>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```



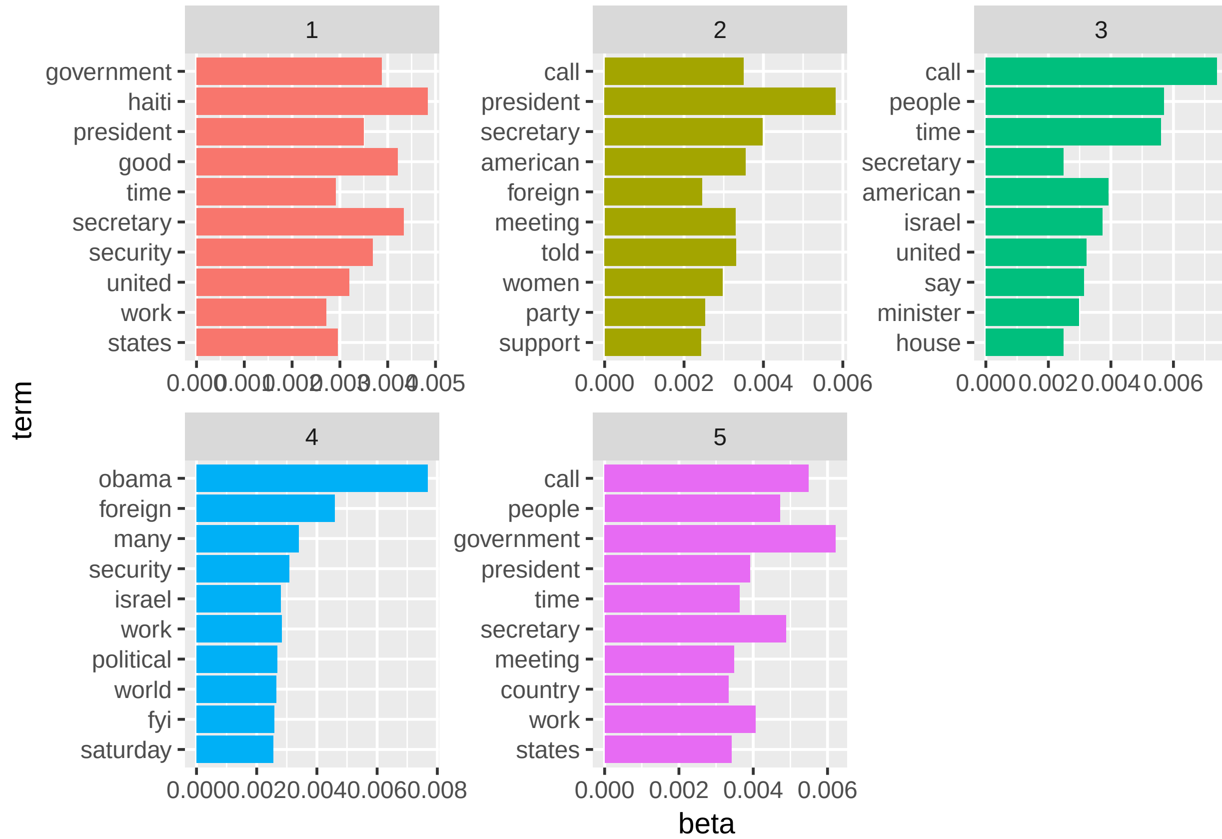

```
# 3 groups


email_nonames_3topic <- LDA(dtm_nonames, k=3 )

topics3 <- tidy(email_nonames_3topic, matrix= "beta")

top_terms <- topics3%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```



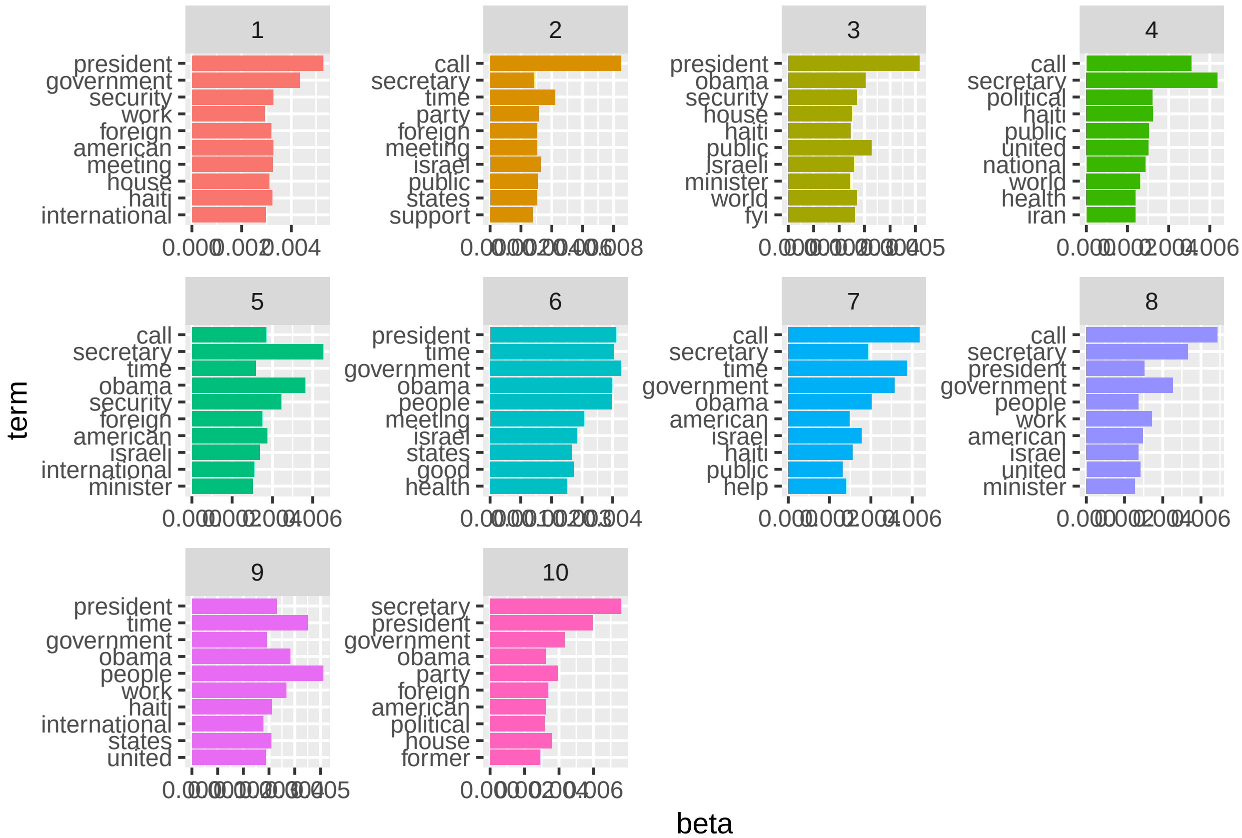

```r
# 4 groups


email_nonames_4topic <- LDA(dtm_nonames, k=4 )

topics4 <- tidy(email_nonames_4topic, matrix= "beta")

top_terms <- topics4%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```




```r
# 5 groups


email_nonames_5topic <- LDA(dtm_nonames, k=5 )

topics5 <- tidy(email_nonames_5topic, matrix= "beta")



top_terms <- topics5%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```

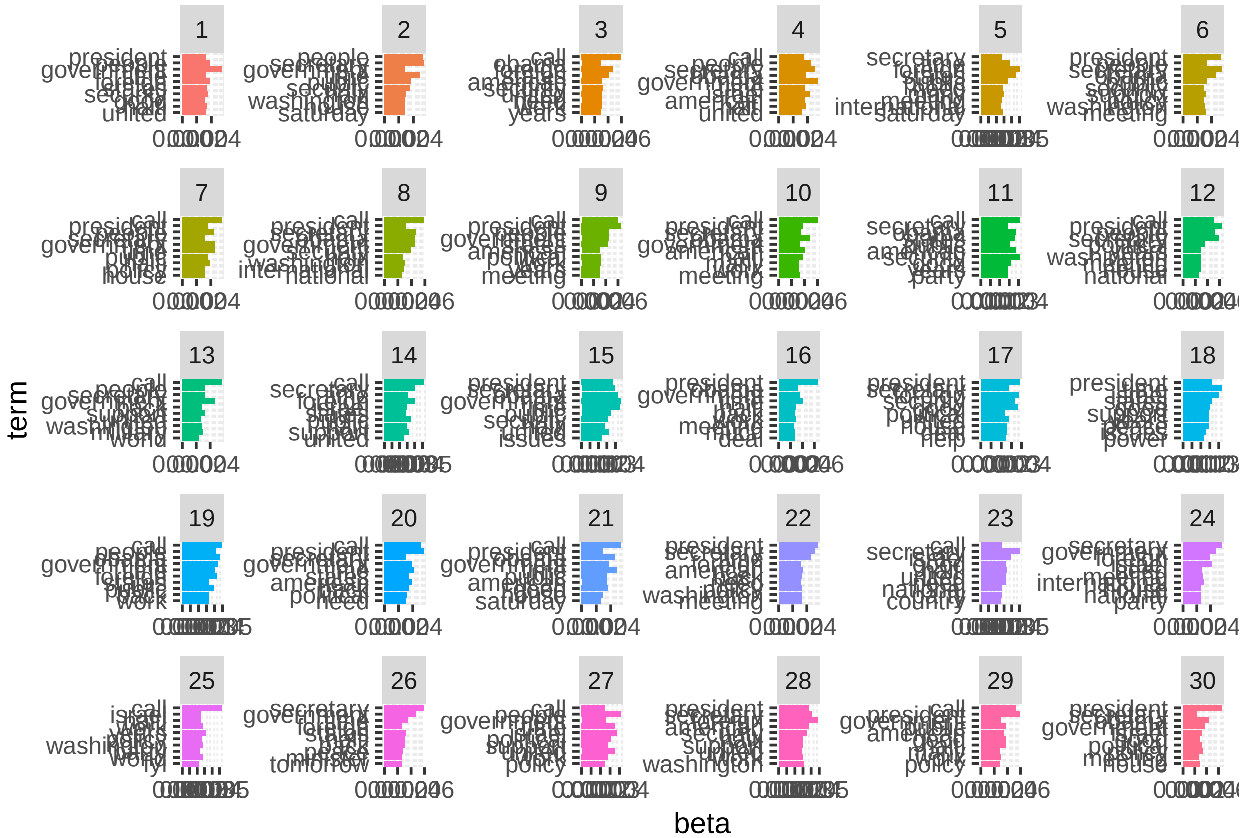



```r
# 10

email_nonames_10topic <- LDA(dtm_nonames, k=10)

topics10 <- tidy(email_nonames_10topic, matrix= "beta")



top_terms <- topics10%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```




```r
# 20

email_nonames_20topic <- LDA(dtm_nonames, k=20)

topics20 <- tidy(email_nonames_20topic, matrix= "beta")



top_terms <- topics20%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```


```r
# 30

email_nonames_30topic <- LDA(dtm_nonames, k=30)

topics30 <- tidy(email_nonames_30topic, matrix= "beta")



top_terms <- topics30%>%
  group_by(topic)%>%
  top_n(10, beta) %>%
  ungroup()%>%
  arrange(topic, -beta)


top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```


term
beta